Universiti Tun Hussein Onn Malaysia

UNIVERSITI TUN HUSSEIN ONN MALAYSIA

FINAL EXAMINATION
SEMESTER II
SESSION 2023/2024

COURSE NAME	:	DATABASE
COURSE CODE	:	BIC 21404
PROGRAMME CODE	:	BIP / BIS / BIW / BIM
EXAMINATION DATE	:	JULY 2024
DURATION	:	3 HOURS
INSTRUCTIONS	:	1. ANSWER ALL QUESTIONS 2. THIS FINAL EXAMINATION IS CONDUCTED VIA ☐ Open book ☒ Closed book 3. STUDENTS ARE **PROHIBITED** TO CONSULT THEIR OWN MATERIAL OR ANY EXTERNAL RESOURCES DURING THE EXAMINATION CONDUCTED VIA CLOSED BOOK

THIS QUESTION PAPER CONSISTS OF **EIGHT (8)** PAGES

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PART A

Q1 Which of the following statements is a feature of a database?

(a) No-backup for the data stored.

(b) User interface provided.

(c) Lack of authentication.

(d) Store data in multiple locations.

(1 mark)

Q2 Which of the following statements is **NOT** a function of a database?

(a) Managing stored data

(b) Manipulating data

(c) Security for stored data

(d) Analysing code

(1 mark)

Q3 Which of the following statements is **CORRECT** to describe normalization in database management?

(a) Normalization is the process of converting unstructured data into a structured format for efficient storage and retrieval

(b) Normalization is a technique used to minimize redundancy and dependency in database design, leading to a more organized and efficient database schema

(c) Normalization involves encrypting sensitive data to ensure its security and prevent unauthorized access

(d) Normalization is the process of optimizing database performance by indexing frequently accessed data fields

(1 mark)

Q4 Which of the following terms is the **CORRECT** format of data stored in DBMS?

(a) Image

(b) Text

(c) Table

(d) Graph

(1 mark)

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Q5 Which phase of database design primarily focuses on defining entities, attributes, and relationships without considering implementation details?

(a) Conceptual Design

(b) Logical Design

(c) Physical Design

(d) Implementation Design

(1 mark)

Q6 Which types of DBMS permits the management of distributed databases and makes distribution transparent to users?

(a) Cloud DBMS

(b) Distributed DBMS

(c) Object-Oriented DBMS

(d) Relational DBMS

(1 mark)

Q7 Which of the following is **NOT** a database type?

(a) Hierarchical

(b) Network

(c) Distributed

(d) Decentralized

(1 mark)

Q8 Which of the following is an **ADVANTAGE** of normalization in database design?

(a) Increased redundancy

(b) Simplified data retrieval

(c) Decreased data integrity

(d) Reduced data consistency

(1 mark)

Q9 Which of the following is **NOT** an example of DBMS?

(a) MySQL

(b) Microsoft Access

(c) IBM DB2

(d) Google

(1 mark)

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Q10 Based on the following, choose the **CORRECT** statement to represent the main concern of Physical Database Design as opposed to Logical Database Design.

(a) Ensuring data integrity through normalization

(b) Designing entity-relationship diagrams

(c) Defining logical views of the database

(d) Optimizing storage and access methods for efficiency

(1 mark)

Q11 What is the primary use of eXtensible Markup Language (XML) in database management?

(a) Data mining

(b) Online Analytical Processing (OLAP) analysis

(c) Semi-structured data representation

(d) Relational Database Management

(1 mark)

Q12 Which of the following is the **CORRECT** type of query that is commonly performed in data warehousing?

(a) Transactional

(b) Predictive

(c) Operational

(d) Analytical

(1 mark)

Q13 Which anomalies can be eliminated in order to remove undesired characteristics?

(a) Insert

(b) Update

(c) Delete

(d) All of the above

(1 mark)

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Q14 When a relation contains an atomic value, it is a _____ relation.

(a) 1NF

(b) 2NF

(c) 3NF

(d) BCNF

(1 mark)

Q15 During transition of a logical database design to a physical database design, which major issue often occurs?

(a) Normalization

(b) Denormalization

(c) Entity-Relationship Modelling

(d) Database Schema

(1 mark)

Q16 Which of the following is designed to optimize the storing and querying of large volumes of multi-dimensional data?

(a) XML

(b) Data Warehousing

(c) OLAP

(d) Data Mining

(1 mark)

Q17 A relation is in _____ if it is in Boyce Codd normal form and does not have any multivalued dependencies.

(a) 1NF

(b) 2NF

(c) 3NF

(d) 4NF

(1 mark)

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Q18 Which of the following is **INCORRECT** to represent a main operation of data mining?

(a) Predictive modelling

(b) Database segmentation

(c) Link analysis

(d) Transaction processing

(1 mark)

Q19 What is cardinality?

(a) The number of entities participating in the relationship

(b) The type of attribute associated with an entity

(c) The uniqueness of an entity's identifier

(d) The data type of an attribute

(1 mark)

Q20 Which of the following is **NOT** an attribute type in the Entity Relationship (ER) model?

(a) Composite attribute

(b) Derived attribute

(c) Primary attribute

(d) Multi-valued attribute

(1 mark)

PART B

Q21 Create an Entity-Relationship Diagram (ERD) using Crow's Foot notation for the business rules in **Figure Q21.1**.

ABC Company is a car rental service that operates in multiple cities. The company wants to develop a database system to manage its rental operations efficiently.

The following business rules have been identified:

- Each car in the fleet is uniquely identified by a registration number.
- A car can be rented multiple times, but each rental transaction must be associated with only one car.
- Customer can rent multiple cars, and each rental transaction must be associated with only one customer.
- Each rental transaction has a unique rental ID and includes information such as rental date, return date, and total rental cost.

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- Some customers may have special membership status, which entitles them to discounts on rentals.
- Customers can rent car based on company that operates in several cities. Each location is identified by a unique location code.

Figure Q21.1

(15 marks)

Q22

(a) List **FIVE (5)** possible threats.

(5 marks)

(b) Based on the answer in **Q22 (a)**, justify **FIVE (5)** best practices for avoiding security threat to databases as they are network accessible.

(10 marks)

Q23 **Table Q23.1** presents the list of dentist and patient appointment data. A patient is given an appointment at a specific time and date with a dentist located during a particular surgery. On each day of patient appointment, a dentist is allocated to a specific surgery for that day.

Table Q23.1 This table shows the lists dentist/patient appointment data.

staffNo	dentistName	patient No	patientName	appointment		surgeryNo
				date	time	
S1011	Tony Smith	P100	Gillian White	12-Sep-01	10:00	S15
S1011	Tony Smith	P105	Jill Bell	12-Sep-01	12:00	S15
S1024	Helen Pearson	P108	Lan MacKay	12-Sep-01	10:00	S10
S1024	Helen Pearson	P108	Lan MacKay	14-Sep-01	14:00	S10
S1032	Robin Plevin	P105	Jill Bell	14-Sep-01	16:30	S15
S1032	Robin Plevin	P110	Johon Walker	15-Sep-01	18:00	S15

(a) Provide **ONE (1)** example for insertion, deletion, and modification anomalies.

(3 marks)

(b) Write the relational schema and draw its dependency diagram. Identify all dependencies, including a transitive dependency.

(4 marks)

(c) Perform data normalization process to meet the Third Normal Form (3NF) requirements.

(8 marks)

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Q24

(a) Compare **TWO (2)** differences between Distributed DBMS and Distributed Processing based on their functionality.

(4 marks)

(b) Compare **TWO (2)** advantages between Centralized DBMS and Distributed DBMS.

(4 marks)

Q25 **Table Q25.1** presents the staff information details.

Table Q25.1: Staff information.

StaffNo	fName	lName	position	sex	DOB	salary	branchNo
SL21	John	White	Manager	M	1-Oct-45	30000	B005
SG37	Ann	Beech	Assistant	F	10-Nov-60	12000	B003

(a) Construct an XML document based on attributes in the table.

(5 marks)

(b) List **TWO (2)** component types of the XML document in **Q25(a).**

(2 marks)

- END OF QUESTIONS -

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